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Supersymmetry cannot be associated with a simple and intuitive change in observational vantage point; shifts in time, in spatial location, in angular orientation, and in velocity of motion exhaust these possibilities. But just as spin is “like rotational motion, with a quantum-mechanical twist,” supersymmetry can be associated with a change in observational vantage point in a “quantum-mechanical extension of space and time.” These quotes are especially important, as the last sentence is only meant to give a rough sense of where supersymmetry fits into the larger framework of symmetry principles.4 Nevertheless, although understanding the origin of supersymmetry is rather subtle, we will focus on one of its primary implications—should the laws of nature incorporate its principles—and this is far easier to grasp.

In the early 1970s, physicists realized that if the universe is supersymmetric, the particles of nature must come in pairs whose respective spins differ by half a unit. Such pairs of particles—regardless of whether they are thought of as pointlike (as in the standard model) or as tiny vibrating loops—are called superpartners. Since matter particles have spin-½ while some of the messenger particles have spin-1, supersymmetry appears to result in a pairing—a partnering—of matter and force particles. As such, it seems like a wonderful unifying concept. The problem comes in the details.

By the mid-1970s, when physicists sought to incorporate supersymmetry into the standard model, they found that none of the known particles—those of Tables 1.1 and 1.2—could be superpartners of one another. Instead, detailed theoretical analysis showed that if the universe incorporates supersymmetry, then every known particle must have an as-yet-undiscovered superpartner particle, whose spin is half a unit less than its known counterpart. For instance, there should be a spin-0 partner of the electron; this hypothetical particle has been named the selectron (a contraction of supersymmetric-electron). The same should also be true for the other matter particles, with, for example, the hypothetical spin-0 superpartners of neutrinos and quarks being called sneutrinos and squarks. Similarly, the force particles should have spin-½ superpartners: For photons there should be